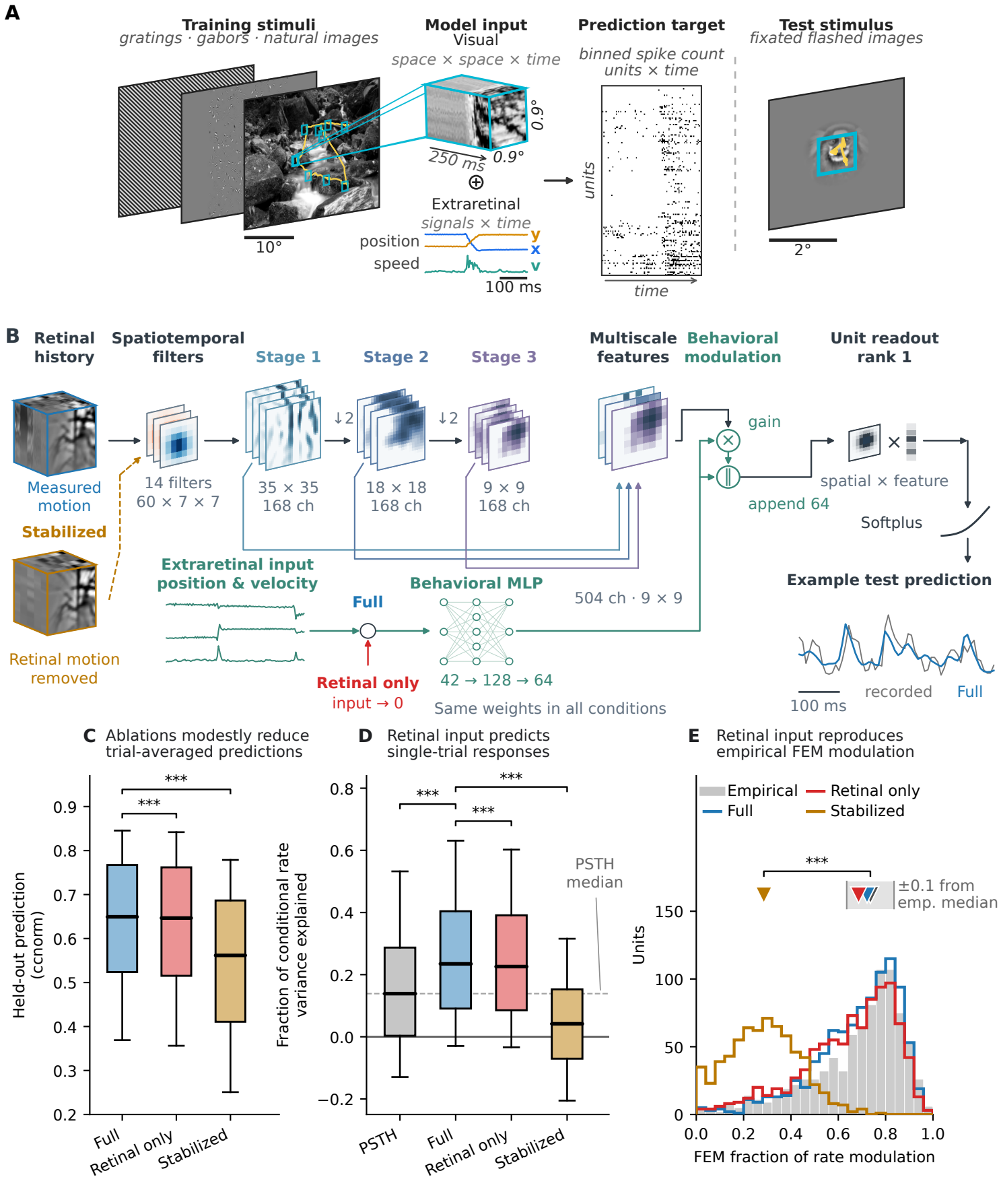

A
Training stimuli
gratings · gabors · natural images
Model input
Visual
space × space × time
0.9°
250 ms
0.9°
Extraretinal
signals × time
position
speed
y
x
v
100 ms
10°
Prediction target
binned spike count
units × time
units
time
Test stimulus
fixated flashed images
2°
B
Retinal history
Spatiotemporal filters
Stage 1
Stage 2
Stage 3
Multiscale features
Behavioral modulation
Unit readout rank 1
Measured motion
Stabilized
Retinal motion removed
14 filters
60 × 7 × 7
35 × 35
168 ch
↓2
18 × 18
168 ch
↓2
9 × 9
168 ch
gain
append 64
spatial × feature
Softplus
Example test prediction
100 ms
recorded
Full
Extraretinal input
position & velocity
Full
Behavioral MLP
504 ch · 9 × 9
Retinal only
input → 0
42 → 128 → 64
Same weights in all conditions
C
Ablations modestly reduce trial-averaged predictions
Held-out prediction (ccnorm)
Full
Retinal only
Stabilized
D
Retinal input predicts single-trial responses
Fraction of conditional rate variance explained
PSTH median
PSTH
Full
Retinal only
Stabilized
E
Retinal input reproduces empirical FEM modulation
Empirical
Retinal only
Full
Stabilized
±0.1 from emp. median
Units
FEM fraction of rate modulation